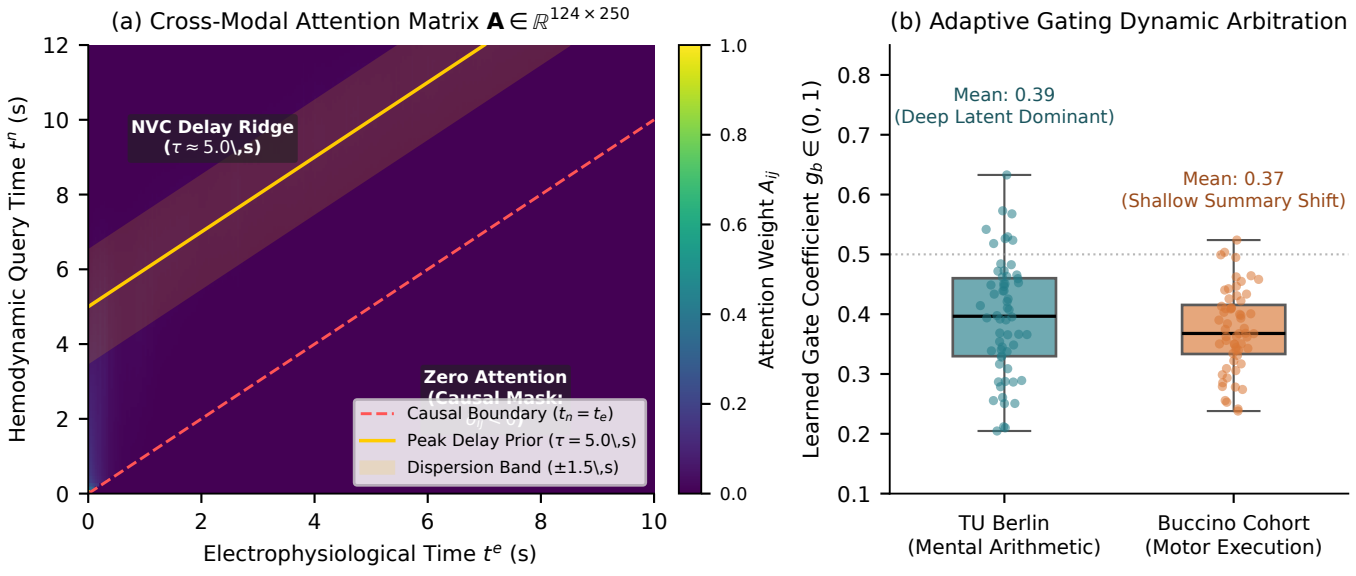

(a) Cross-Modal Attention Matrix $\mathbf{A} \in \mathbb{R}^{124 \times 250}$
NVC Delay Ridge
($\tau \approx 5.0$\,s)
Zero Attention
(Causal Mask:
$t_j < t_i$)
Causal Boundary ($t_n = t_e$)
Peak Delay Prior ($\tau = 5.0$\,s)
Dispersion Band (± 1.5\,s)
Hemodynamic Query Time t^n (s)
Electrophysiological Time t^e (s)
Attention Weight A_{ij}
0
2
4
6
8
10
12
0.0
0.2
0.4
0.6
0.8
1.0
(b) Adaptive Gating Dynamic Arbitration
Learned Gate Coefficient $g_b \in (0, 1)$
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Mean: 0.39
(Deep Latent Dominant)
Mean: 0.37
(Shallow Summary Shift)
TU Berlin
(Mental Arithmetic)
Buccino Cohort
(Motor Execution)